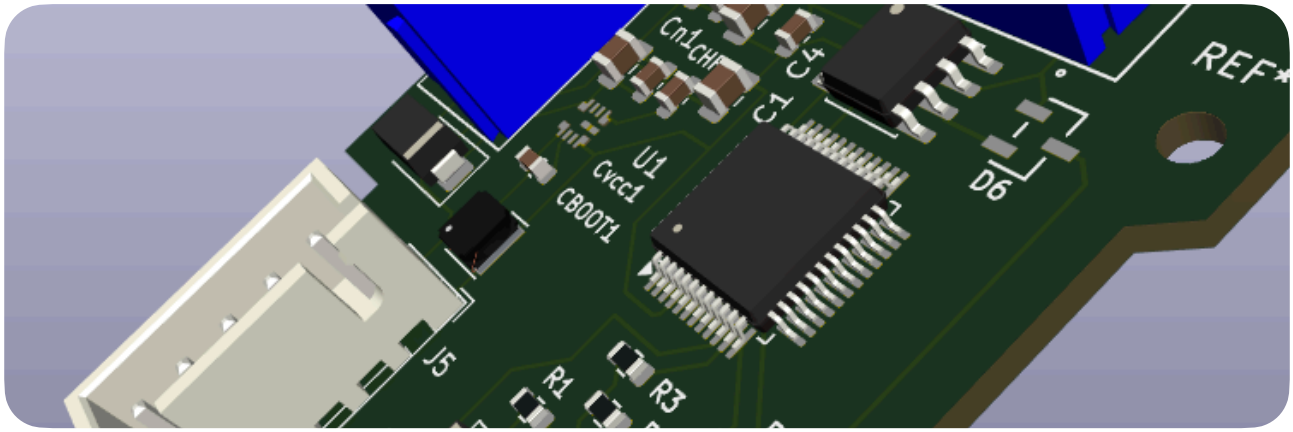


ADC-to-CAN

Analog Measurement Module

Hardware Design & Technical Documentation

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1. Introduction

The project focuses on an analog measurement module designed for automotive applications. The module is powered directly from the vehicle's low-voltage electrical system and measures input voltages, converts the analog signals into digital values, and transmits the measured data to the ECU via the CAN bus.

The ADC-to-CAN module can be used to interface with voltage-output sensors, provided their output signals are compatible with the module's input range. The CAN bus reduces the need for separate signal wiring, supports message prioritisation, and provides good resistance to electromagnetic interference when properly implemented.

The project includes a detailed circuit schematic, a PCB design, mechanical mounting details with a custom-designed bracket, and technical documentation covering the main aspects of the design.

2. Requirements

Requirement	Design target
Microcontroller	STM32 family
Analog inputs	Minimum 2 channels
Input voltage range	0-16V
CAN communication	CAN 2.0, 1 Mbps
Measurement frequency	Target 100hz
Supply voltage	Vehicle LV system, nominal 14.7V
Operating input range	9-16V
Reverse polarity protection	Required
PCB dimensions	Maximum 50x50 mm
PCB technology	2-layer PCB, SMD components

PBC Design Approach

A 2-layer PCB was selected to keep to keep the design relatively simple, compact and cost-effective. Two copper layers provide enough space for routine the signals and power connections while keeping the manufacturing process straightforward. This approach is suitable for a project this size and complexity.

Design Priorities

The main priorities of the project are compact dimensions and reliable operation. The PCB should fit within the required maximum dimensions of 50 x 50 mm, while maintaining enough space for components, connectors and routing. Current size of the PCB is 39 x 49 mm.

Requirements and Verification

The design was prepared based on the requirements provided for the project. Parameters such as the number of analog inputs, supply voltage range and Can communication speed, were included directly in the design. However, certain characteristics, including measurement accuracy, actual CAN communication performance and protection against electromagnetic interference, still require verification on physical hardware.

3. System Architecture

3.1 General System Description

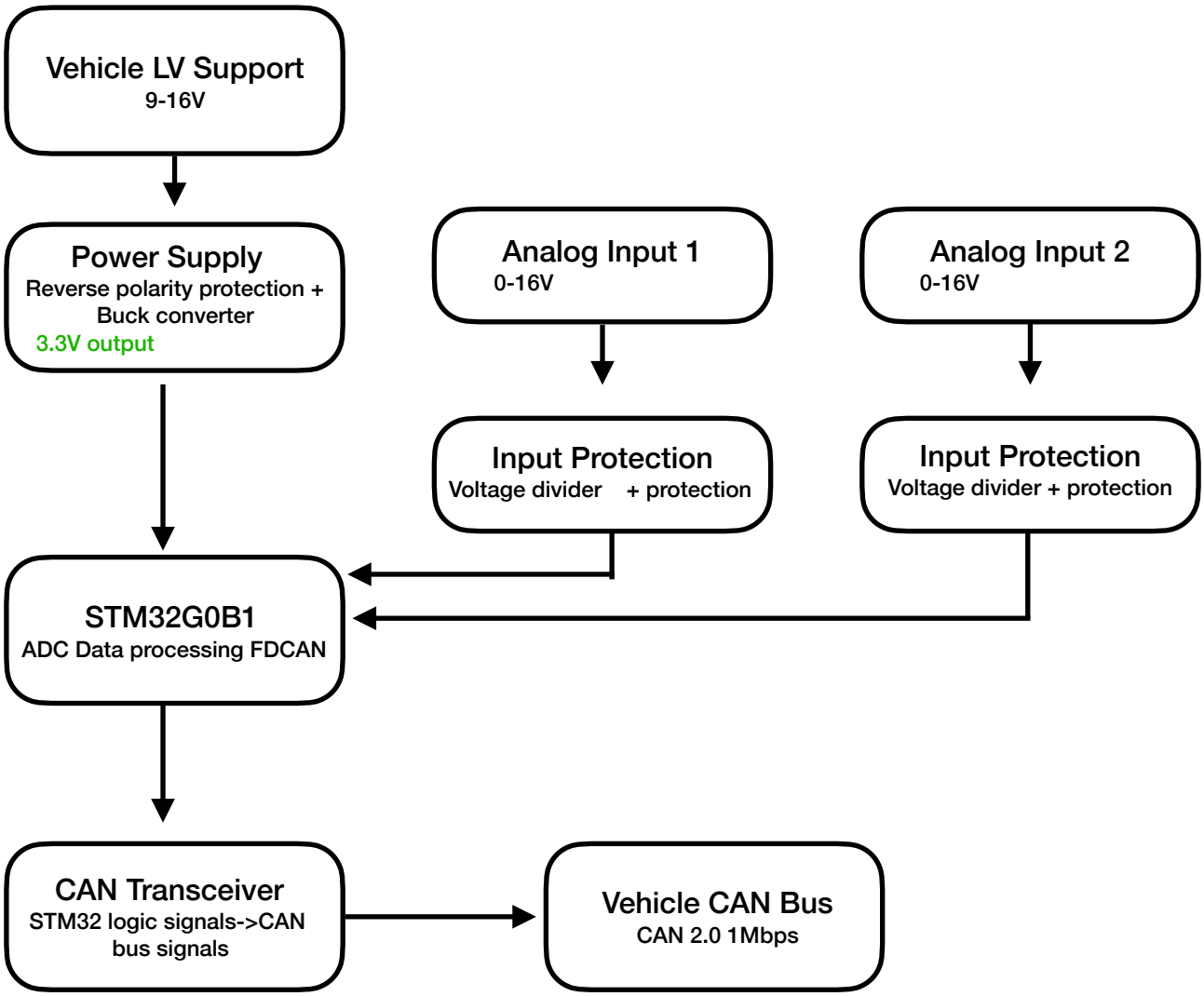
The ADC to CAN module is designed to measure two analog voltage signals and transmit the measured values over a CAN bus.

The module is powered directly from the vehicle low-voltage system. The input voltage is converted to 3.3V, which is used to power the microcontroller and CAN transceiver.

Each analog input passes through a voltage divider and an input protection circuit before reaching the STM32 microcontroller. The microcontroller uses its ADC peripheral to read and convert the signals into digital values.

The measured data is then processed by the firmware and transmitted through the CAN transceiver to the vehicle CAN bus.

3.2 System Block Diagram



Parameter	Value
CAN standard	CAN 2.0
Bitrate	1 Mbps
CAN ID	0x123
Frame type	Standard Data Frame
Payload	4 bytes
Transmission interval	Approximately 10 ms

4.Hardware Design

This section describes the main hardware components used in the ADC to CAN module. The design consists of a microcontroller, a power supply circuit, two analog circuits, a CAN interface and a programming connector.

3.1 Microcontroller



The main processing unit of the module is the **STM32B0B1 microcontroller(U2)**. It was selected because it provides the required ADC functionality and an integrated FDCAN controller.

The microcontroller operates with 3.3V supply voltage. Its ADC peripheral is used to read the two analog input channels connected to pins PA0 and PA1. The measured values are processed by firmware and prepared for transmission over the CAN bus.

Decoupling capacitors are connected to the microcontroller’s power supply to help maintain a stable supply voltage and reduce unwanted noise. The microcontroller is also connected to the external CAN transceiver through its FDCAN communication signals.

3.2 Power Supply

The module is designed to perform powered directly from the vehicle’s low-voltage electrical system. The input voltage range specified for the design is 9-16V.

The power inout circuit includes an **SS34 Schottky diode (D1)** connected in series with the supply line. It provides reverse polarity protection by blocking current when the input supply is connected with the wrong polarity.

The main voltage regulator is **LMR36503-Q1 buck converter(U1)**. It converts the input voltage to a regulated 3.3V output, which is used to power the microcontroller and the CAN transceiver.

Additional capacitors and an inductor are used as part of the converter circuit. These components support voltage regulation and help reduce variations in the output voltage.

3.3 Analog Input Circuit

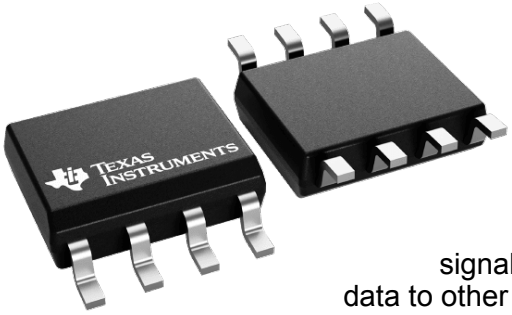
The module has two analog input channels connected through connectors J2 and J3. Each channel is designed to measure an input voltage within the specified range of 0-16V.

Since the STM32 microcontroller operates at 3.3V, the input voltage must be reduced before it reaches the ADC. This is dine using a resistor voltage divider. The divider scales the input signal to a level suitable for the microcontroller's analog input.

Each channel also includes an **SM4T21AY TVS diode (D4 and D5)**, which is intended to help protect the input circuit against voltage disturbances. Additional resistors and capacitors are included in the input circuits to support signal conditioning and filtering.

The conditioned signals are connected to the microcontroller's ADC inputs PA0 and PA1. Allowing both channels to be measured independently.

3.4 CAN Interface



Communication with the vehicle's CAN bus is handled by the microcontroller's integrated FDCAN peripheral and an external **TCAN332 transceiver (U3)**.

The transceiver connects the microcontroller's logic-level communication signals to the differential CAN bus signals. It allows the module to transmit measurement data to other devices connected to the vehicle network.

The CAN interface is designed for **CAN 2.0 communication at 1Mbps**. AN **ESD2CAN24DBZRQ1 protection device (D6)** is included in the CAN signal path to help protect the interface against electrostatic discharge.

The CAN signals are routed to the J4 connector, which provides the physical connection to the vehicle's CAN network.

3.5 Programming Interface

Connector J5 provides access to the microcontroller's **Serial Wire Debug (SWD)** interface.

It is used to program the STM32 microcontroller and debug the firmware during development. The connector provides access to the SWDIO and SWCLK signals, as well as the reset signal, 3.3V supply and ground.

This interface makes it possible to upload firmware and inspect the microcontroller's operation without requiring a separate programming circuit on the PCB.

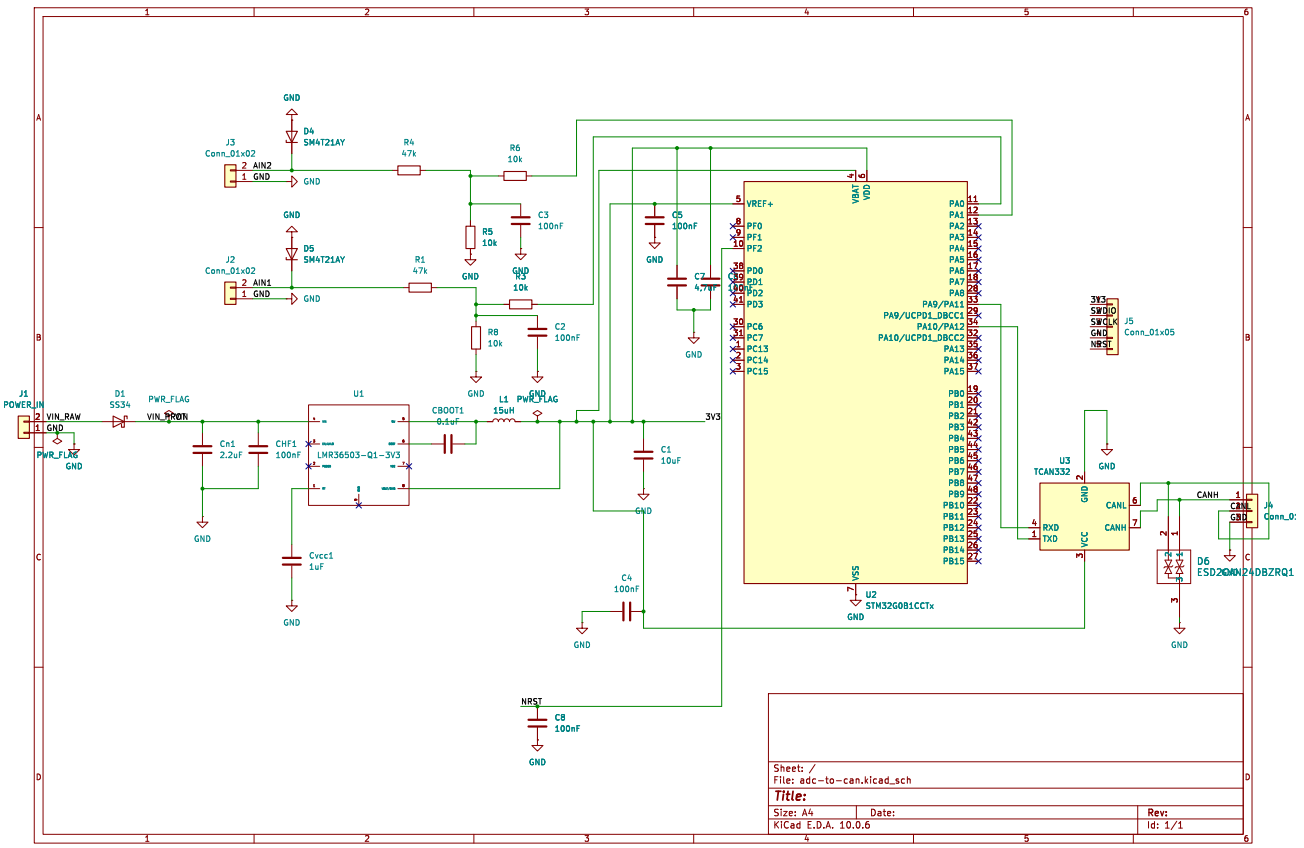
5.6 Connector selection

Screw terminal connectors were selected for the main power input and CAN bus connection. These connections may need to remain secure during vehicle operation, so screw terminals provide a more secure mechanical connection and reduce the risk of wires becoming accidentally disconnected due to vibration.

For analog inputs and programming interface, plug in connectors were chosen instead. These connections are intended to make assembly, disconnection and maintenance easier. The also allow the input signals and programming interface to be connected or disconnected without having to unscrew individual wires.

5. PCB Design

5.1 Schematic

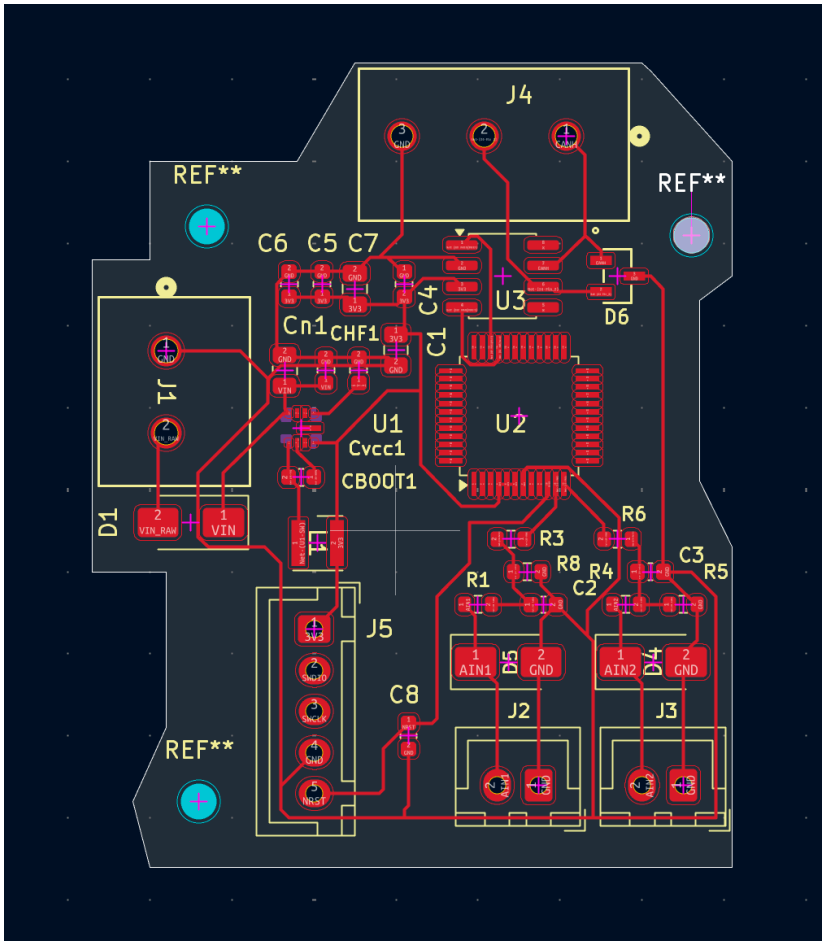


The electrical schematic was created in KiCad and represents the connections between all the components of the ADC to CAN module. It includes the power supply circuit, two analog input channels, the STM32 microcontroller, the CAN interface and the SWD programming connector.

The schematic was divided into functional sections to make the design easier to understand and work with. Power gorund and signal connections were assigned appropriate net labels to keep the circuit organized.

After completing the schematic, an Electrical Rules Check (ERC) was performed. The final schematic passed the check with **0 errors and 0 warnings**.

5.2 PBC Layout

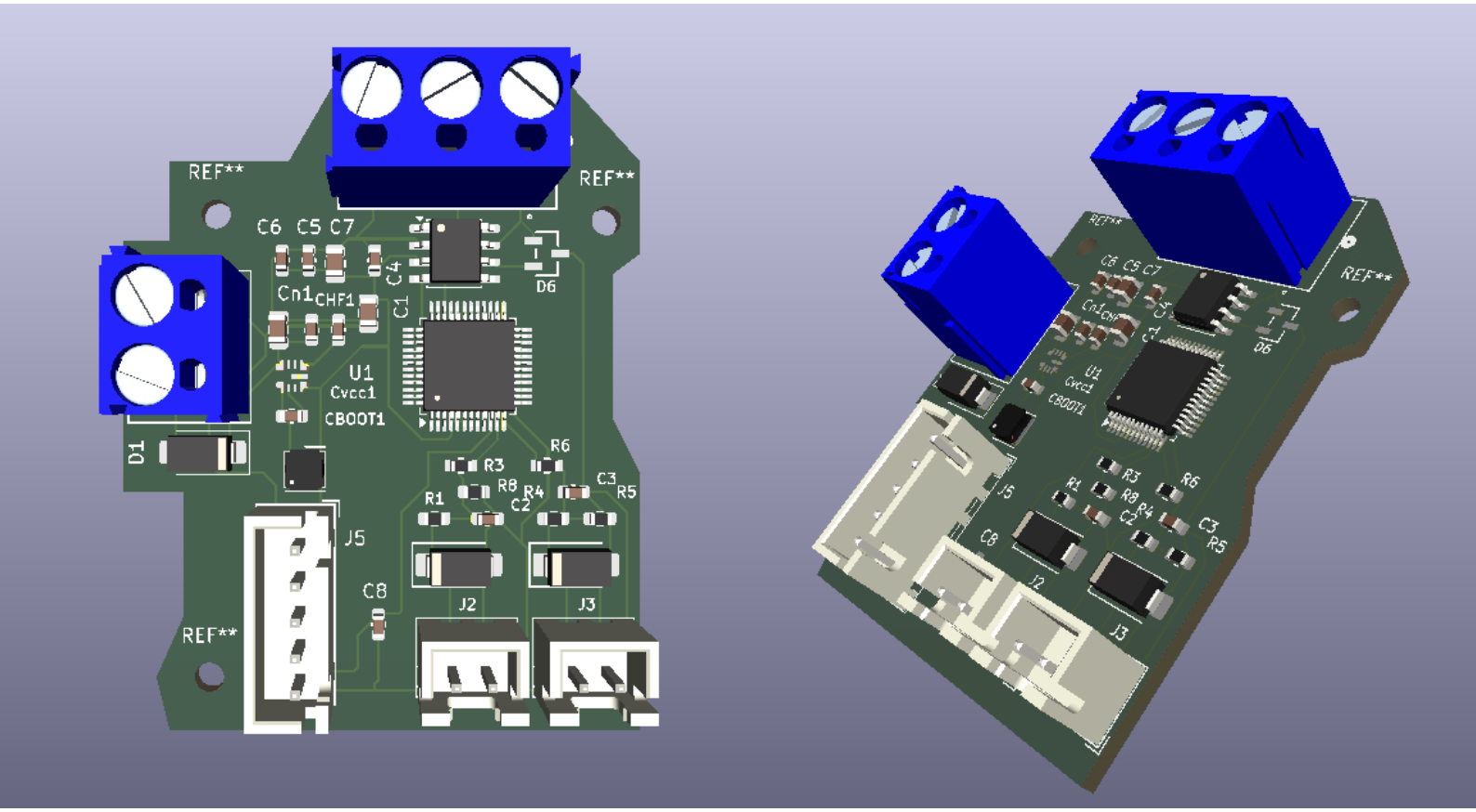


The PCB layout was designed in KiCad as a 2-layer board, with surface-mount components used throughout the design, except for connectors where through-hole or other suitable mounting options were considered.

During component placement the main goal was to keep the board compact while maintaining a clear arrangement of functional sections. Components and analog input circuits, were places close to each other and near the board edges for the practicality.

The layout was checked using the Design Rules Check (DRC). The final result contained **0 errors and 1 warning**. The warning concerned a modified component footprint, which should be checked against the manufacturer's data sheet before PCB manufacturing.

5.3 3D Model


REF**
REF**
C6 C5 C7
C4
C1
Cn1
CHF1
D6
U1
Cvcc1
CBOOT1
D1
R6
R3
R8
R4
C2
C3
R5
R1
J5
C8
J2
J3
REF**

6. Firmware Overview

6.1 Overview

The firmware is designed to get two analog voltage measurements using the ADC peripheral of the STM32G0B1 microcontroller and transmit the processed measurements over a CAN bus.

The application is developed in C using STM32 HAL library and STM32CubeIDE. Peripheral initialisation code is generated using STM32CubeMX.

6.2 Analog Data Acquisition

The ADC1 peripheral is configured for 12-bit resolution and a sequence of two conversions. The first conversion uses ADC **channel 0**, while the second one uses ADC **channel 1**. These channels are intended to correspond to the **PA0** and **PA1** pins.

Each conversion is checked for successful completion before the measured value is read. The raw ADC results are stored as unsigned 32-bit integers. Measurements are processed only when both conversions are valid and completed.

6.3 Voltage Calculation

The measured voltage is reconstructed from the ADC result using ADC resolution, the assumed 3.3V reference voltage and voltage divider ratio.

The conversion is performed using:

$$V_{in} = \frac{V_{raw}}{4095} \times 3.3 \times 5.7$$

The result is converted to millivolts and stored as a 16-bit unsigned integer.

$$V_{in} \times 1000$$

The calculation assumes a 3.3V ADC reference and nominal voltage divider ratio of 5.7.

6.4 Can Communication

The FDCAN peripheral is configured to transmit measurement data using Classical CAN frames with standard 11-bit identifier of 0x123.

Each frame contains four data bytes representing the two voltage measurements. Each measurement is an unsigned 16-bit value In millivolts, with the most significant byte transmitted first.

Byte	Contents
0	Channel 1 - most significant byte
1	Channel 1 - least significant byte
2	Channel 2 - most significant byte
3	Channel 2 - least significant byte

The firmware attempts to enqueue a Can frame after successful of both measurements. A transmission error queue error increments an error counter.

6.5 Execution Cycle

The main loop performs the following operations:

1. Start ADC conversions.
2. Read the two ADC results.
3. Verify if both conversions are complete.
4. Convert the raw results into millivolts.
5. Encode both measurements into a four byte CAN payload.
6. Attempt to enqueue the frame for transmission.
7. Wait for around 10ms before next iteration.

The delay is intended to produce nominal update interval of approximately 10ms. The actual transmission rate depends on execution time and Can controller operation.

6.6 Verification Status

The firmware currently build successfully without any errors or warnings. This confirms compilation only: the firmware has not been yet verified on physical hardware.

7. Verification and Testing

The table below contains each test and its verification status at the moment:

Test/Verification	Method	Status
Schematic verification	KiCad ERC	Done
PBC design verification	KiCad DRC	Done
Firmware compilation	STM32CubeIDE	Done
ADC measurement accuracy	Comparison wth reference voltage	Pending
CAN communication	CAN loopback test	Done
Power supply output	Measure 3.3V output	Pending
Reverse polarity protection	Electrical test	Pending

- ERC - schematic is clear, with 0 errors and 0 warnings.
- DRC - PCB layout passes the test with 0 errors and 1 warning(the warning refers to the changes footprint of the component due to it being invalid from the manufacturer)
- Firmware Compilation - 0 errors, 0 warnings.

8. Bill of Materials

Ref.	Qty	Value	Package	Manufacturer	Datasheet
C1	1	10uF	CL31B106KOHVPN#	Samsung	https://product.samsungsem.com/resources/file/product-catalog/MLCC_Automotive_2401.pdf?utm_source=chatgpt.com
C2-C6,C8,CHF1	7	100nF	CL31B104KAPWPN#	Samsung	-||-

Ref.	Qty	Value	Package	Manufacturer	Datasheet
C7	1	4,7uF	CL31B475KOHVPN#	Samsung	-||-
CBOOT1	1	100nF	CL31B104KAPWPN#	Samsung	-||-
Cn1	1	2.2uF	CL31B225KOHVPN#	Samsung	-||-
Cvcc1	1	1uF	CL31B105KCHVPN#	Samsung	-||-
D1	1	SS34		Onsemi	https://www.onsemi.com/pdf/datasheet/ss39-d.pdf
D4,D5	2	SM4T21AY		STMicroelectronics	https://www.st.com/resource/en/datasheet/cd00283395.pdf
D6	1	ESD2CAN24DBZRQ1		Texas Instruments	https://www.ti.com/lit/ds/symlink/esd2can24-q1.pdf?ts=1793166872850
J1	1	POWER_IN		Würth Elektronik	https://www.we-online.com/catalog/datasheet/691137710002.pdf
J2,J3	2	Conn_01x02		JST	https://www.lcsc.com/datasheet/C158012.pdf
J4	1	Conn_01x03		Würth Elektronik	https://www.we-online.com/components/products/datasheet/691101710003.pdf

Ref.	Qty	Value	Package	Manufacturer	Datasheet
J5	1	Conn_01x05		JST	https://www.mouser.pl/datasheet/3/1070/1/eXH.pdf
L1	1	15uH		Coilcraft	https://www.mouser.pl/datasheet/3/995/1/lps5030.pdf
R1,R4	2	47k		Parasonic Industry	https://industrial.panasonic.com/ww/products/pt/anti-sulfurated-chip-resistors/models/ERJS03F4702V
R3,R5,R6,R8	4	10k		Parasonic Industry	https://industrial.panasonic.com/cdbs/www-data/pdf/RDA0000/AOA0000C304.pdf
U1	1	LMR36503RS3ERPERQ1	VQFN-HR	Texas Instruments	https://www.ti.com/lit/ds/symlink/lmr36503e-q1.pdf?ts=1793173241025&ref_url=https%253A%252F%252Fwww.google.com
U2	1	STM32G0B1CCTx	LQFT-48	STMicroelectronics	https://www.st.com/resource/en/datasheet/stm32g0b1cc.pdf
U3	1	TCAN332	SOT-23	Texas Instruments	http://www.ti.com/lit/ds/symlink/tcan337.pdf